

WEEK 5  
CD LAB  
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230905560  
CSE -B  
Roll- 64

**Questions:**

**Q1. Count the number of vowels and consonants in the given input.**

**CODE:**

```
%{  
#include<stdio.h>  
int vow=0;  
int cons=0;  
%}  
  
%%  
[aeiouAEIOU] {vow++;}  
[a-zA-Z] {cons++;}  
\n {return 0;}  
. {}  
%%
```

```

int yywrap() { return 1;}

int main()
{

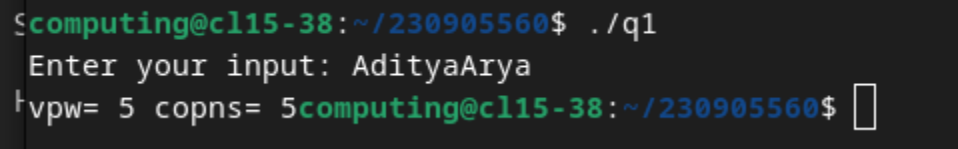
char stat[20];
printf("Enter your input: ");

//scanf("%s",stat);

yylex();      //reads ther input(no need of scanf)
printf("vpw= %d copns= %d",vow,cons);
}

```

**OUTPUT:**



```

computing@cl15-38:~/230905560$ ./q1
Enter your input: AdityaArya
vpw= 5 copns= 5computing@cl15-38:~/230905560$ 

```

**Q2. Count the number of words, characters, blanks and lines in each text.**

**CODE:**

```

%{
#include<stdio.h>

int w=1;
int c=0;
int bl=0;
int line=1;
%}

```

%%

\n {line++;w++;}

" " {bl++;w++;}

\t {bl++;}

. {c++;}

%%

int yywrap() { return 1;}

int main()

{

char stat[20];

printf("Enter your input: ");

//scanf("%s",stat);

yylex(); //reads the input(no need of scanf)

printf("\nwords= %d characters= %d blanks=%d lines=%d",w,c,bl,line);

}

**OUTPUT:**

```
vpw= 5 copns= 5computing@cl15-38:~/230905560$ ./q2
Enter your input: aditya aryan sahu
words= 3 characters= 15 blanks=2 lines=1computing@cl1
```

**Q3. Find the number of positive integer, negative integer, positive floating point number and negative floating point number.**

## CODE:

```
%{  
#include<stdio.h>  
int p=0,n=0,pf=0,nf=0;  
%}  
  
%%  
[-][0-9]+ {n++;}  
[0-9]+ {p++;}  
"-"[0-9]+\.[0-9]+ {nf++;}  
[0-9]+\.[0-9]+ {pf++;}  
%%  
  
int yywrap() { return 1;}  
int main()  
{  
  
char stat[20];  
printf("Enter your input: ");  
//scanf("%s",stat);  
yylex();      //reads ther input(no need of scanf)  
printf("\npositive= %d negative= %d pos_float=%d neg_float=%d",p,n,pf,nf);  
}
```

OUTPUT:

```
Enter your input: 1 -1 -1.2 2.4 6
positive= 2 negative= 1 pos_float=1 neg_float=1
```

**Q4. Given a input C file, replace all scanf with READ and printf with WRITE statements also find the number of scanf and printf in the file.**

**CODE:**

```
%{
#include<stdio.h>

int s_count=0;
int p_count=0;
}%

%%

"scanf" { s_count++; printf("READ"); }
"printf" { p_count++; printf("WRITE"); }
. { printf("%s", yytext); }
\n { printf("%s", yytext); }
%%

int yywrap() { return 1; }

int main()
{
```

```
FILE *fp;

printf("Enter file name: ");

char str[100];

scanf("%s",str);

fp = fopen(str, "r");

if(fp==NULL)

{

    printf("Error haha");

    return 1;

}

yyin=fp;

yylex();


printf("Number of scanf: %d\n", s_count);

printf("Number of printf: %d\n", p_count);


return 0;

}
```

INPUT FILE:

```
1 int main(){
2 int a;
3 scanf("%d", &a);
4
5 printf("trail code\n");
6 printf(("vale uf a \n"));
7
8 //hahahahah
9
10
11 }
```

OUTPUT:

```
computing@cl15-38:~/230905560$ ./q4
Enter file name: test.c
int main(){
int a;
READ("%d", &a);

WRITE("trail code\n");
WRITE(("vale uf a \n"));

//hahahahah

}
Number of scanf: 1
Number of printf: 2
```

Q5. That changes a number from decimal to hexadecimal notation.

CODE:

```
%{
#include <stdio.h>
#include <stdlib.h>
%}
```

%%

[0-9]+ {

int decimal = atoi(yytext); //str to int conv

printf("%X", decimal);

}

. { printf("%s", yytext); }

\n { printf("%s", yytext); }

%%

int yywrap() {

return 1;

}

int main() {

printf("Enter decimal numbers:\n");

yylex();

return 0;

}

OUTPUT:

```
computing@cl15-38:~/230905560$ ./q5
Enter decimal numbers:
123
7B
45
2D
```

**Q6. Convert uppercase characters to lowercase characters of C file excluding the characters present in the comment.**

**CODE:**

```
%{
#include<stdio.h>

int s_count=0; i
nt p_count=0;

%}

%%

[a-z] {printf("%c",yytext[0]-32); }
[A-Z] {printf("%c",yytext[0]+32); }

/* ""/[a-z]* { char c; while ((c = input()) != '\n' CC c != 1); printf("%s",yytext); }*/

""/"" { printf("%s", yytext);

    char c;

    while ((c = input()) != '\n' CC c != 0)

    {

        putchar(c);

    }
```

```

        if (c == '\n')

            putchar('\n'); }

. { printf("%s", yytext); }

\n { printf("%s", yytext); }

%%

int yywrap() { return 1; }

int main() {

FILE *fp;

printf("Enter file name: ");

char str[100];

scanf("%s", str);

fp = fopen(str, "r");

if(fp==NULL)

{ printf("Error haha");

return 1; }


yyin=fp; yylex();

//printf("Number of scanf: %d\n", s_count);


// printf("Number of printf: %d\n", p_count);

return 0;

}

```

INPUT FILE:

```
1 int main(){
2 int a;
3 scanf("%d", &a);
4
5 printf("trail code\n");
6 printf(("vale uf a \n"));
7
8 //hahahahah
9
10
11 }
```

OUTPUT:

```
computing@cl15-38:~/230905560$ ./q6
Enter file name: test.c
INT MAIN(){
INT A;
SCANF("%D", &A);

PRINTF("TRAIL CODE\n");
PRINTF(("VALE UF A \N"));
C
//hahahahah

}
```